

In []:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

# Load the data into a pandas DataFrame
file_path='C:\sales_data.csv'

df = pd.read_csv(file_path)

# Display the first few rows of the DataFrame
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# Summary statistics
print(df.describe())

# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print(product_summary)
```

In []:

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# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)

# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)

# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

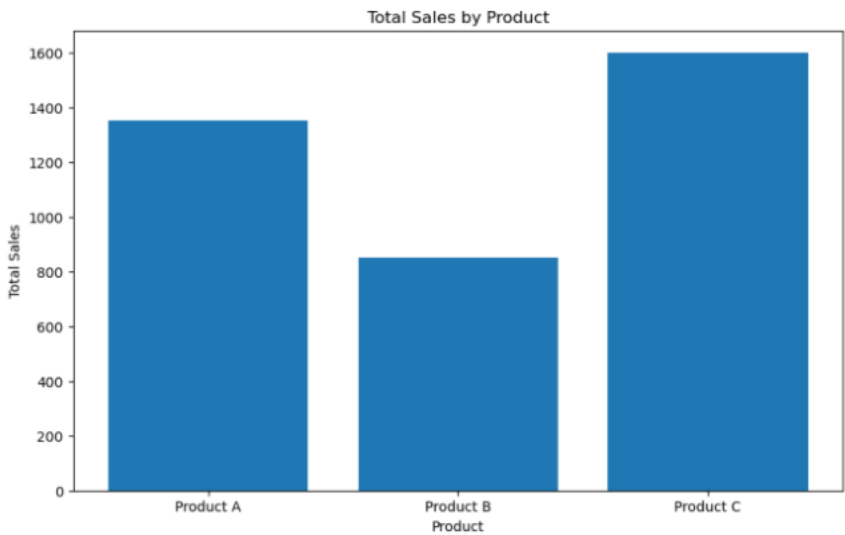
	Date	Product	Sales	Quantity	Region
0	2023-01-01	Product A	200	4	North
1	2023-01-02	Product B	150	3	South
2	2023-01-03	Product A	220	5	North
3	2023-01-04	Product C	300	6	East
4	2023-01-05	Product B	180	4	West

Date 0
Product 0
Sales 0
Quantity 0
Region 0

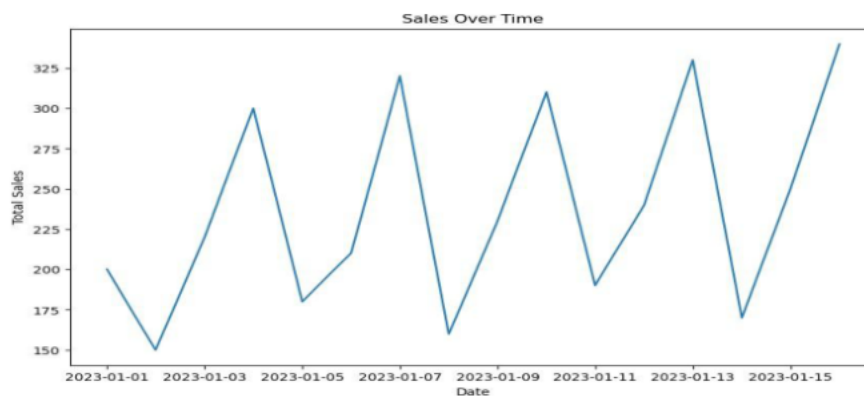
dtype: int64

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36



In[]:



```
Product  Product A  Product B  Product C
Region
East      0         0        1600
North    1350         0         0
South      0        480         0
West      0        370         0
```

```
      Sales  Quantity
Sales    1.000000  0.944922
Quantity 0.944922  1.000000
```

C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated . In a future version, it will default to False. Select only valid columns or specify the value of numeric_only to silence this warning.

```
correlation_matrix = df.corr()
```

